Biochemistry Cumulative Examination
February 6, 2004

1. How would you describe the "primary", "secondary", "tertiary", and "quaternary" structure of proteins? (20%)

2. Give two examples where protein quaternary structure is important for biological function. (20%)

3. In examining crystal structural determination of proteins one often see the terms: resolution, R factor, phase problem and electron density map. Define the meaning and the significance of these terms in the procedure of X-ray structural determination. (20%)

4. In NMR structural determination of proteins, we often see other terms: stable isotope (^{15}N and ^{13}C) labeling of protein, sequence-specific resonance assignment, NOE and torsion angle constraints, distance-geometry calculation, and the ensemble of structures. Explain the significance of these terms in the procedure of NMR structural determination. (20%)

5. Comparing to NMR method, what are the advantages and disadvantages of single crystal X-ray techniques in terms of 3D structural determination of proteins? (20%)